

Problem 3

Problem 3

Problem 1 True or False:

(1) If $f(x) = (x - 2)^x$, then $f'(x) = x(x - 2)^{x-1}$.

Explanation. False. Any time that you have a function of x raised to a function of x , in order to compute the derivative you need to use logarithmic differentiation (or something equivalent).

Correct derivative of f :

$$\begin{aligned} f(x) &= (x - 2)^x \implies f(x) = e^{x \ln(x-2)} \\ \implies f'(x) &= e^{x \ln(x-2)} \cdot \left(1 \cdot \ln(x-2) + x \cdot \frac{1}{x-2} \right) \\ \implies f'(x) &= (x - 2)^x \cdot \left(\ln(x-2) + \frac{x}{x-2} \right) \end{aligned}$$

(2) If $f(x) = (3x)^x$, then $f'(x) = (3x)^x \ln(3x)$.

Explanation. False. Same as part (1).

Correct derivative of f :

$$\begin{aligned} f(x) &= (3x)^x \implies f(x) = e^{x \cdot \ln(3x)} \\ \implies f'(x) &= e^{x \cdot \ln(3x)} \left(1 \cdot \ln(3x) + x \cdot \frac{3}{3x} \right) \\ \implies f'(x) &= (3x)^x (\ln(3x) + 1) \end{aligned}$$